

Erdős Problem #885

EP-885

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For integer $n \geq 1$ we define the factor difference set of n by $D(n) = \{|a - b| : n = ab\}$. Is it true that, for every $k \geq 1$, there exist integers $N_1 < \dots < N_k$ such that $|\cap_i D(N_i)| \geq k$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>